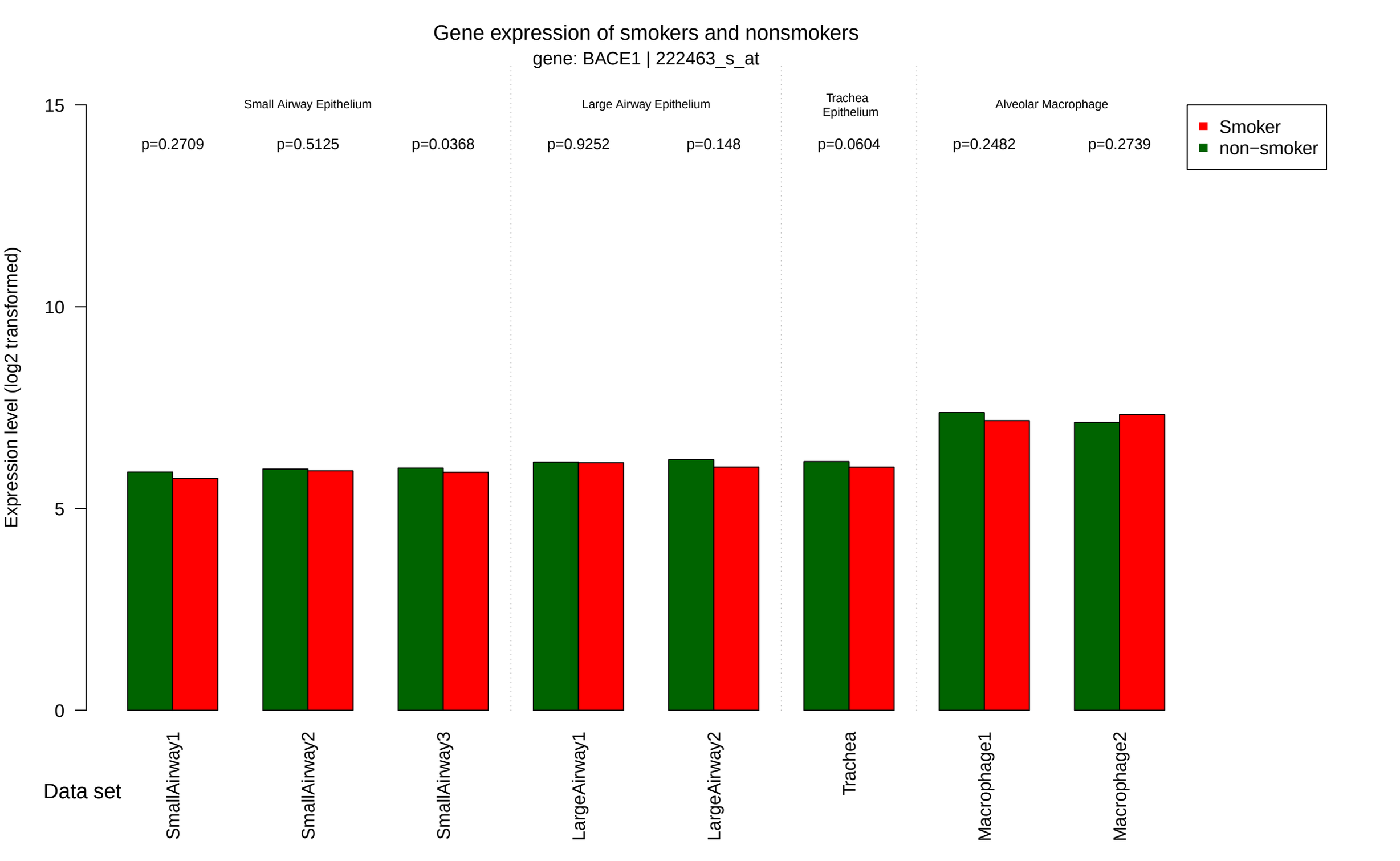
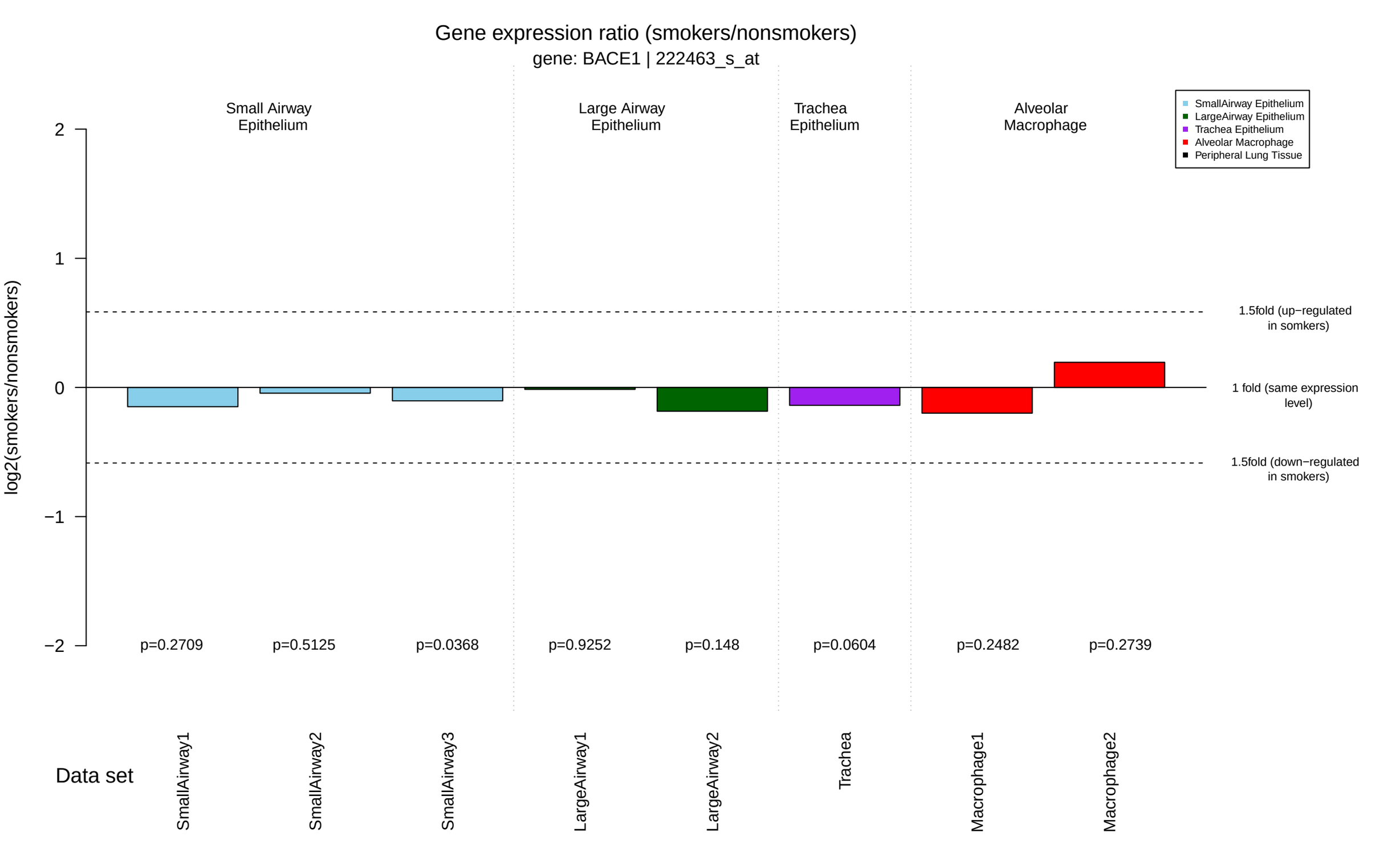
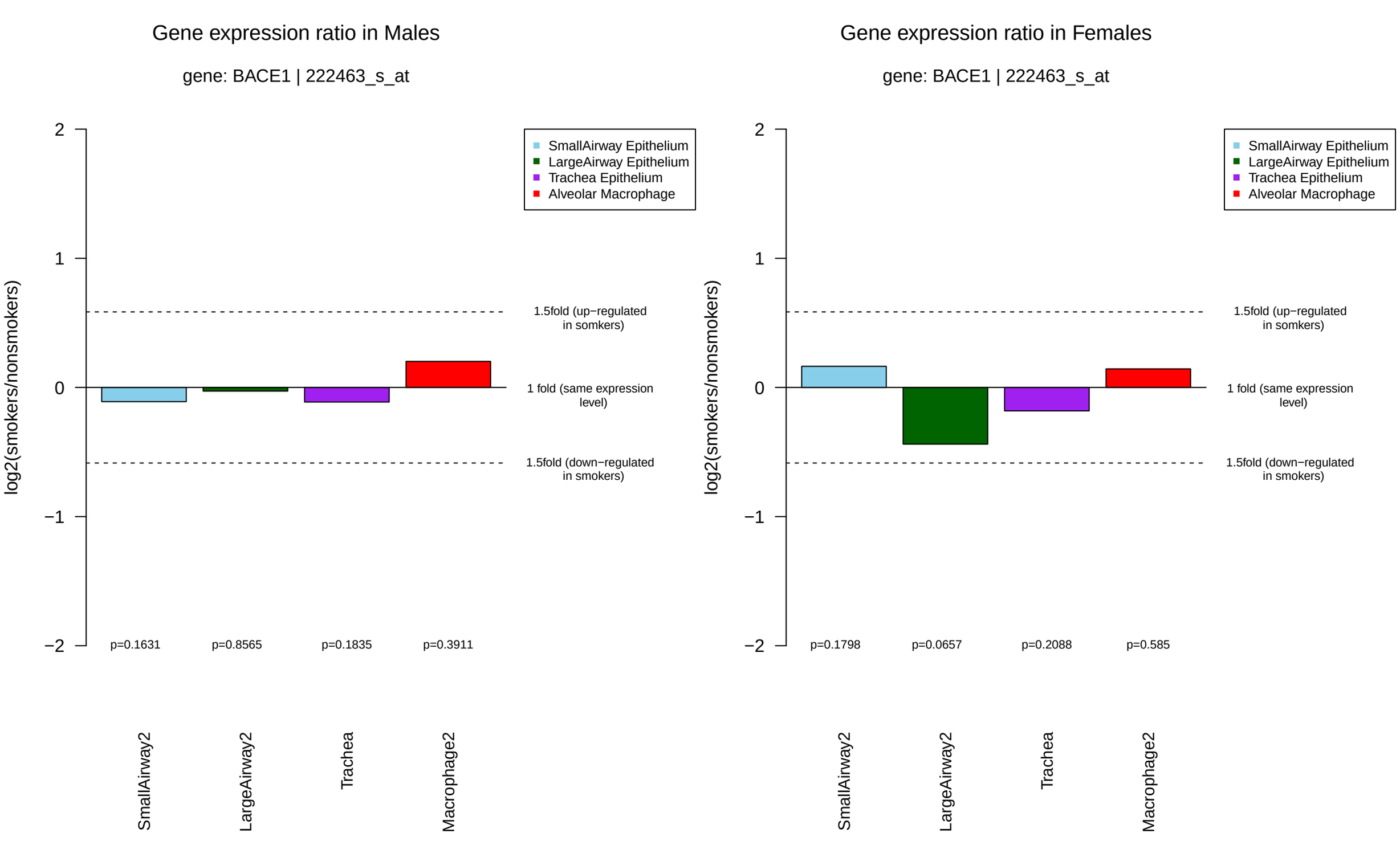
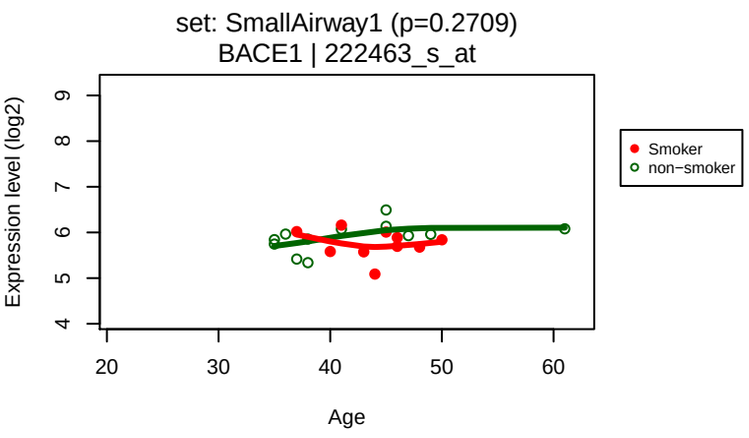
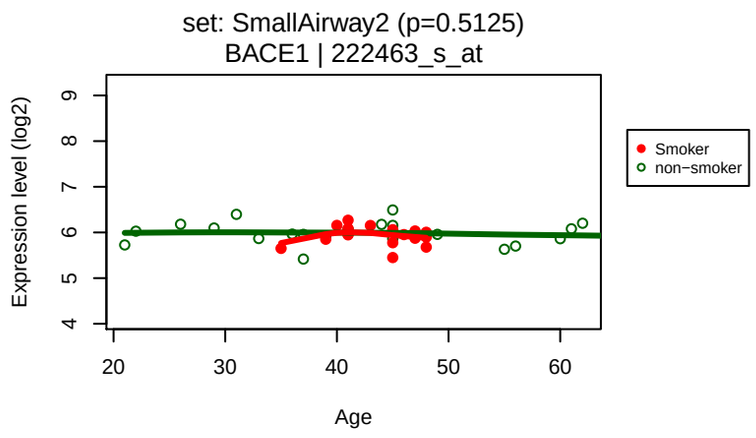
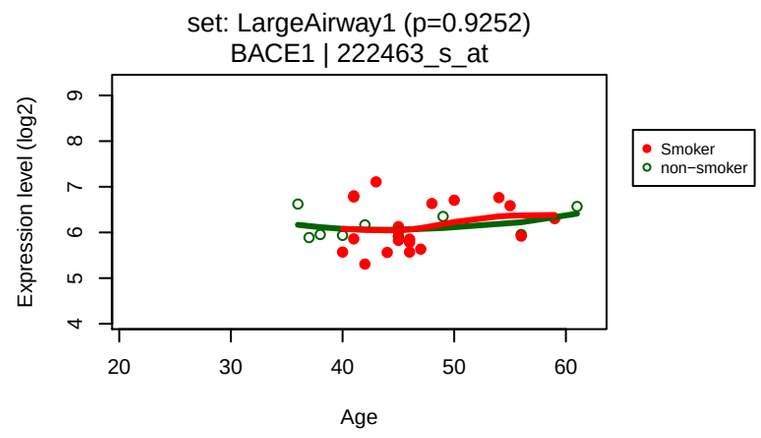

Gene expression of smokers and nonsmokers
gene: BACE1 | 222463_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2709
p=0.5125
p=0.0368
p=0.9252
p=0.148
p=0.0604
p=0.2482
p=0.2739
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

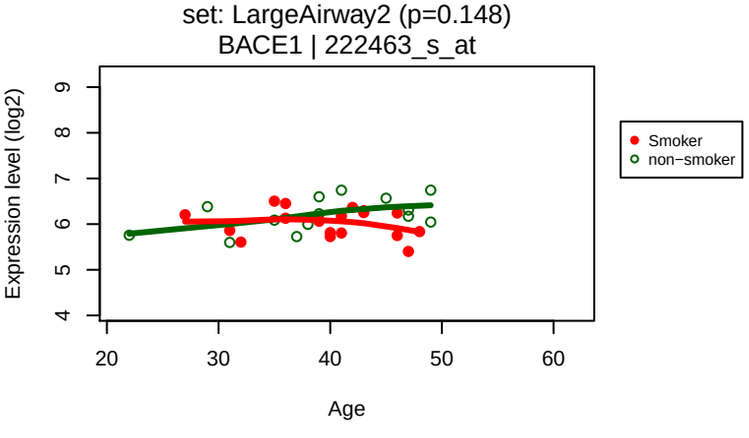

Gene expression ratio (smokers/nonsmokers)
gene: BACE1 | 222463_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2709
p=0.5125
p=0.0368
p=0.9252
p=0.148
p=0.0604
p=0.2482
p=0.2739
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

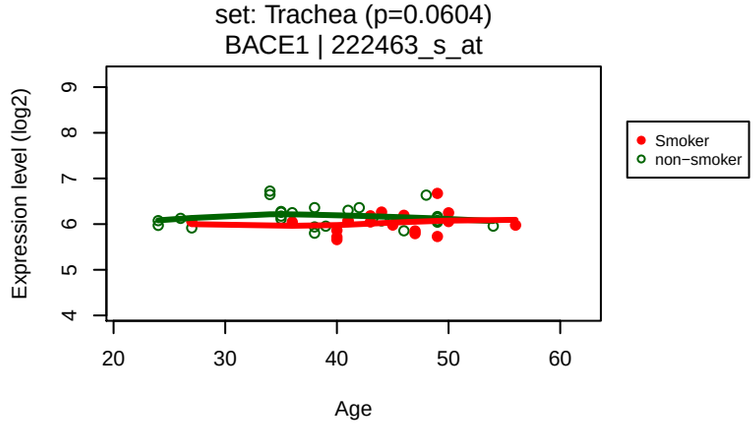

Gene expression ratio in Males
gene: BACE1 | 222463_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1631
p=0.8565
p=0.1835
p=0.3911
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BACE1 | 222463_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1798
p=0.0657
p=0.2088
p=0.585
SmallAirway2
LargeAirway2
Trachea
Macrophage2

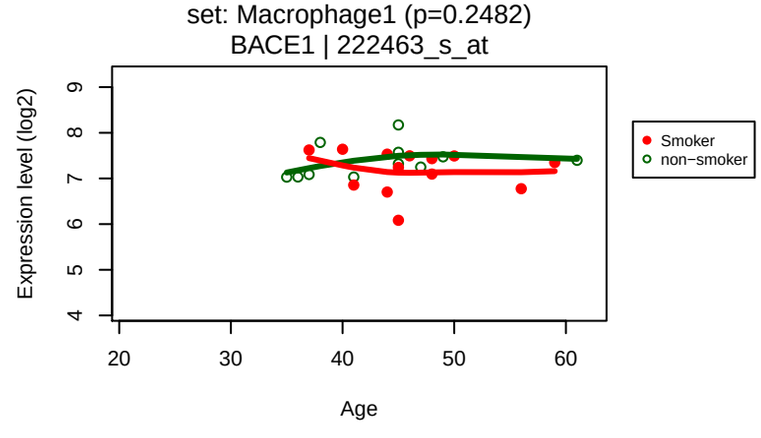

set: SmallAirway1 (p=0.2709)
BACE1 | 222463_s_at
Expression level (log2)
Age
Smoker
non−smoker

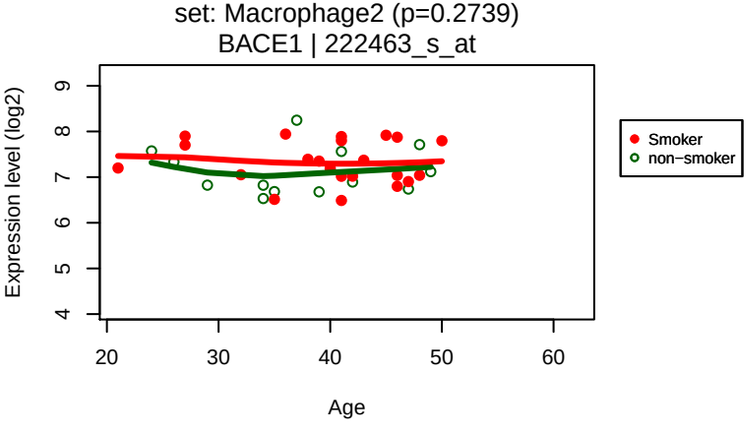

set: SmallAirway2 (p=0.5125)
BACE1 | 222463_s_at
Expression level (log2)
Age
Smoker
non−smoker

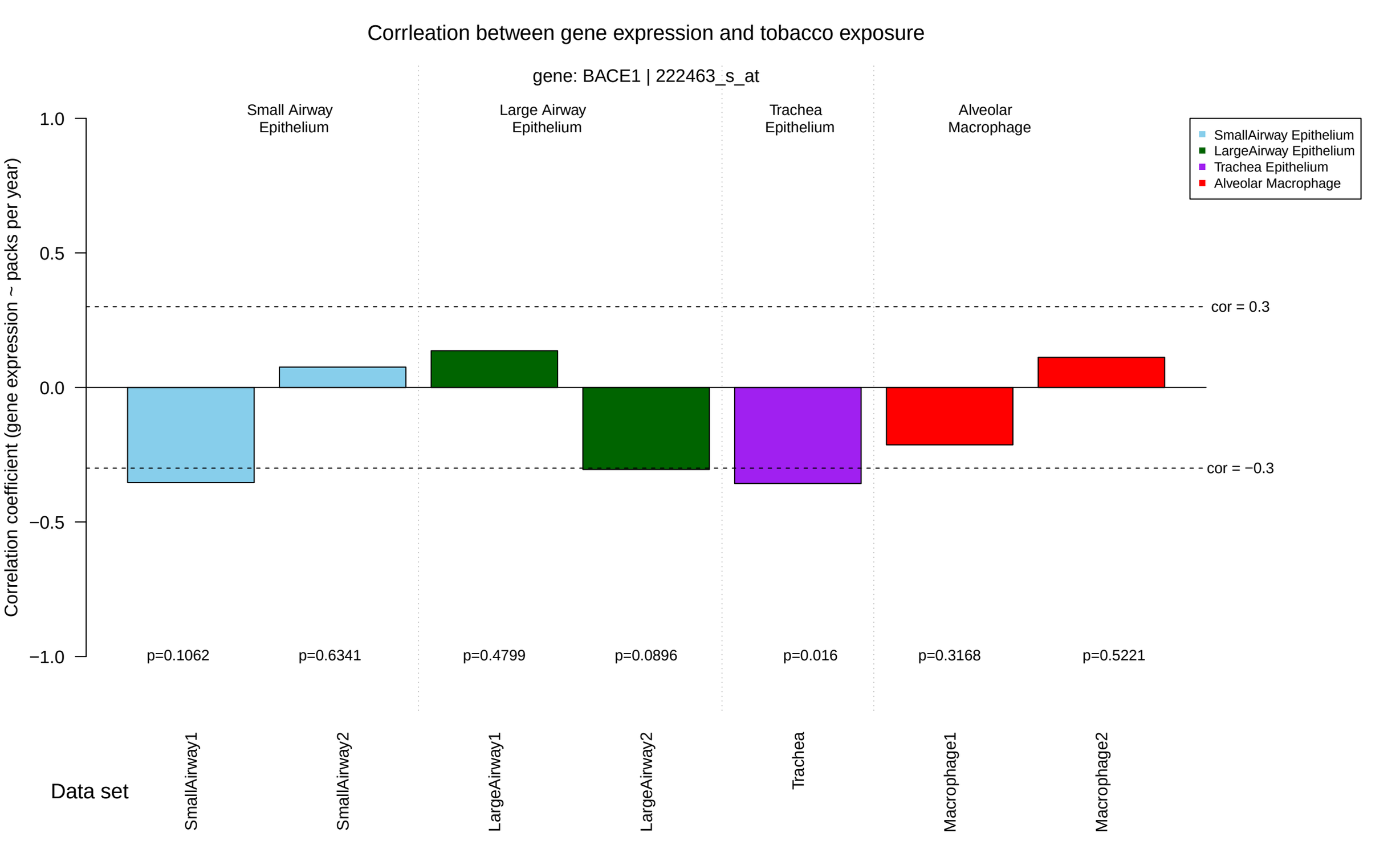

set: LargeAirway1 (p=0.9252)
BACE1 | 222463_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.148)
BACE1 | 222463_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0604)
BACE1 | 222463_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2482)
BACE1 | 222463_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2739)
BACE1 | 222463_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BACE1 | 222463_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.1062
p=0.6341
p=0.4799
p=0.0896
p=0.016
p=0.3168
p=0.5221
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2